

Keras Functional API — Multi-Output Models Explained

1. Overview

The Keras Functional API allows you to build complex models with multiple inputs and outputs. Unlike the Sequential API, it supports shared layers and non-linear architectures, making it suitable for multi-task learning.

2. Supported Scenarios

- Classification (binary or multi-class)
- Regression (single or multiple continuous outputs)
- Mixed tasks (e.g., classification + regression)

3. Example Scenarios Case 1 — Two Regressions

Predict both house price and rent value from the same input.

```
inp = Input(shape=(10,)) x = Dense(64, activation='relu')(inp) x = Dense(32, activation='relu')(x)
price_output = Dense(1, name='price')(x) rent_output = Dense(1, name='rent')(x) model =
Model(inputs=inp, outputs=[price_output, rent_output]) model.compile(optimizer='adam',
loss={'price': 'mse', 'rent': 'mse'})
```

Case 2 — Two Classifications

Predict gender (binary) and age group (multi-class).

```
gender_output = Dense(1, activation='sigmoid', name='gender')(x) age_output = Dense(5,
activation='softmax', name='age_group')(x) model = Model(inputs=inp, outputs=[gender_output,
age_output]) model.compile(optimizer='adam', loss={'gender': 'binary_crossentropy', 'age_group':
'categorical_crossentropy'}, metrics={'gender': 'accuracy', 'age_group': 'accuracy'})
```

Case 3 — Mixed (Classification + Regression)

Predict whether a house will sell (classification) and its price (regression).

```
class_output = Dense(1, activation='sigmoid', name='will_sell')(x) price_output = Dense(1,
activation='linear', name='price')(x) model = Model(inputs=inp, outputs=[class_output,
price_output]) model.compile(optimizer='adam', loss={'will_sell': 'binary_crossentropy', 'price':
'mse'})
```

4. Training

During training, you pass labels for each output:

```
model.fit(X, {'price': y_price, 'rent': y_rent}, epochs=10)
```

Keras automatically optimizes the total loss, which can be a weighted sum if `loss_weights` are provided.

5. Summary Table

Scenario	Supported?	Example Output Layer
Single classification	■	<code>Dense(num_classes, 'softmax')</code>
Single regression	■	<code>Dense(1, 'linear')</code>
Two regressions	■	<code>Two Dense(1, 'linear')</code>
Two classifications	■	<code>Two Dense(..., 'softmax' or 'sigmoid')</code>
Mixed (classification + regression)	■	<code>Dense(softmax) + Dense(linear)</code>

6. Key Takeaway

The Keras Functional API is ideal for any architecture that requires multiple outputs, shared layers, or hybrid tasks — making it a go-to choice for advanced deep learning designs.